MIL-S-80086D
30 June 1986
SUPERSEDING
MIL-S-80086C
5 September 1980

MILITARY SPECIFICATION

SHEARING MACHINES, METAL SQUARING, HYDRAULIC DRIVE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers hydraulically driven metal squaring shearing machines with equipment and accessories as described herein.

1.2 Classification. Shearing machines shall be of one type and of the sizes listed in table I. The size to be furnished shall be as specified (see 6.2.1). Size is determined by the thickness and width of workpiece capacity.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 3445

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

2.1.2 Other Government documents. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

29 CFR, Chapter XVII, Part 1910 - Occupational Safety and Health Standards.

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B11.4 - Safety Requirements for the Construction, Care, and Use of Shears.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

Source: https://assist.dla.mil -- Downloaded: 202610-23T04:37Z
Check the source to verify that this is the currentversion before use.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

JOINT INDUSTRIAL COUNCIL (JIC)

JIC H-1 - Hydraulic Standards for Industrial Equipment and General Purpose Machine Tools.

(Application for copies should be addressed to the Joint Industrial Council, 7901 Westpark Drive, McLean, VA 22101.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The machine shall be new and one of the manufacturer's current models capable of shearing and squaring material in accordance with the requirements specified herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Dimensional systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one dimensional system is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all dimensions, measurements, sizes, and capacities are given in US units. These dimensions may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

Source: https://assist.dla.mil -- Downloaded: 202610-23T04:37Z
Check the source to verify that this is the current version before use.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, front or back gauge indicators, pressure gauges, temperature indicators, and other similar devices. When only one system of graduations is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.1.1 Dual calibrations. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the SI system of measurements.

3.2.2 Reclaimed materials. The shearing machine may contain reclaimed materials provided such materials will not jeopardize the intended use and performance of the machine. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The shearing machine and its components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Controls. All operating controls shall be located conveniently to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the shearing machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the applicable requirements contained in ANSI B11.4 and 29 CFR, Chapter XVII, Part 1910. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. The machine shall be so designed and constructed that, under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

Source: https://assist.dla.mil -- Downloaded: 202610-23T04:37Z
Check the source to verify that this is the current version before use.

3.2.9 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible. When specified (see 6.2.1), an automatic lubrication system shall be furnished.

3.2.10 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The shearing machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine at its maximum rated capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects and mismatching. Processes such as welding, peening, plugging, or filling with solders or paste shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the shearing machine shall be painted in accordance with the manufacturer's standard commercial practice and color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28.

3.3.7 Dials. All dials shall be permanently and legibly engraved or etched on a non-glare background. Digital indicators with direct readings which do not require any calculation may be used in lieu of etched dials.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.

3.3.8 Gears. All gears shall be constructed from a material suitable for the intended purpose. All drive gears shall have the hardness, surface finish, and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure, or premature deterioration and wear.

3.3.9 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The shearing machine shall consist of not less than the following major components.

3.4.1 Frame and base. The frame assembly shall support all other components of the machine at a level of rigidity required to perform the shearing operations specified herein. The frame shall include two end housings, a table, and an upper cross member or members joined together to form a single unit. Unless otherwise specified the end housing shall be the gap type, with a recessed gap of not less than 6 inches from the shearing knives. When specified (see 6.2.1), the frame shall be the closed end type or, when specified, the frame shall be either the gap or closed end type. The end housings shall directly support the guides. The frame members shall be constructed of either a cast or a welded fabrication. The members shall be joined either with a bolted interlocking construction or as a stress relieved welded assembly. The base of the end housings shall have provisions for securely anchoring the machine to a flat mounting foundation.

3.4.2 Table. The table shall be constructed of either cast iron or steel plate, with a surface finish of 125 microinches or less, and flat to within +0.005 inch per linear foot. The table shall securely support the lower shear knife and shall be provided with hand slots to facilitate workpiece handling. The surface of the table shall also be provided with slots, as required, for attaching workpiece positioning gauges specified herein. The table shall be provided with a means of adjustment for obtaining and maintaining clearance alignment between the upper and lower shear knives. When specified (see 6.2.1), the table top of machines rated less than 1/2 inch capacity shall be provided with a ball transfer arrangement. The table top of machines rated 1/2 inch or greater shall be provided with a ball transfer arrangement consisting of steel balls of not less than 3/4 inch diameter. The balls shall be mounted in self-cleaning cups and arranged in a staggered pattern throughout the table top.

3.4.3 Hydraulic system. The machine shall be powered by a hydraulic system that will accomplish the maximum shearing capacity as specified herein and to operate the workpiece hold down cylinders. When specified (see 6.2.1), the hydraulic system shall conform to JIC H-1 Hydraulic Standards. The shear knife shall be reciprocated by a ram which is attached to one or more hydraulic piston and cylinder assemblies. The hydraulic system shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.

be provided with a power unit including an electric motor driven hydraulic pump, fluid reservoir, valves, piping, cylinders, and fluid control devices. Fluid control devices shall be provided to relieve and prevent loss of hydraulic pressure that could cause damage to the machine. The hydraulic system shall be provided with connecting tubing or hoses capable of withstanding not less than 150 percent of the rated working pressure of the system.

3.4.3.1 Ram hydraulic cylinder drive. Unless otherwise specified, the machine shall have either one or more hydraulic units with a mechanical linkage or two direct acting hydraulic cylinder units for driving the ram. When only one type drive is acceptable, the particular drive required shall be as specified (see 6.2.1). The cylinder unit shall be either solid one piece, tie rod, or bolted type construction. Each ram hydraulic cylinder unit shall be of the double acting piston type and the piston rod of each cylinder shall be sealed against leakage. The ram end of the piston rod on the two direct acting cylinder units shall be provided with ball sockets or self aligning joints with provisions for adjustment for wear. Hydraulic cylinder units with mechanical linkage shall also have provisions for adjustment for wear.

3.4.3.2 Hydraulic holddown system. The machine shall be provided with a hydraulic holddown system capable of securing any size workpiece up to and including the rated capacity of the machine. The holddown units shall have a centerline spacing of not more than 12 inches across the full workpiece opening of the machine. The holddown units shall apply uniform pressure, preventing any workpiece movement that would impede the required shearing performance specified herein. The holddown units shall automatically secure the workpiece to the table during the ram downstroke and release the workpiece on the return stroke. When specified (see 6.2.1), the holddown system shall be provided with an independent control which will allow the operator to advance the holddown units without shearing the workpiece.

3.4.3.3 Reservoir. The machine shall have a fluid reservior with a capacity of not less than 200 percent of the required operating supply of hydraulic fluid for the system. The reservoir shall be provided with a fluid level gauge, a fluid filler strainer, and a cleanable filter system. The filter system shall be capable of removing particles 10 microns in size from the hydraulic fluid.

3.4.4 Ram. The ram shall have the strength and rigidity required to hold and support the moveable shear knife in the shearing action. The ram shall be guided through the shearing stroke by ram guides arranged to hold the shear knife in alignment. The ram guides shall guide the ram on a vertical shearing plane of not more than two angular degrees off the shearing plane of the stationary knife.

3.4.5 Shear knives. The machine shall be provided with shock resisting tool steel shear knives. The shear knives shall be designed so that they can be rotated to bring a new cutting edge into operating position and shall be hardened and ground to comply with the manufacturer's standard practice. The cross sectional dimensions of the shear knives shall be not less than the dimensions specified in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.

TABLE I. Sizes and capacities.

Machine size	Rated capacity			Strokes per minute	Shear knives cross section (inches)
	Workpiece thickness (inches)	Workpiece width (feet)	At rake angle setting of in. per foot (not more than)		
1/4 x 6	1/4	6	3/8	20	3/4 x 2-1/4
1/4 x 8	1/4	8	3/8	20	3/4 x 2-1/4
1/4 x 10	1/4	10	3/8	15	3/4 x 2-1/4
1/4 x 12	1/4	12	3/8	13	3/4 x 2-1/4
1/4 x 14	1/4	14	3/8	13	1 x 2-5/8
1/4 x 16	1/4	16	3/8	13	1 x 2-5/8
1/4 x 18	1/4	18	3/8	12	1 x 2-5/8
1/4 x 20	1/4	20	3/8	12	1 x 2-5/8
3/8 x 6	3/8	6	1/2	17	1 x 3
3/8 x 8	3/8	8	1/2	13	1 x 3
3/8 x 10	3/8	10	1/2	11	1 x 3
3/8 x 12	3/8	12	1/2	10	1 x 3
3/8 x 14	3/8	14	1/2	8	1 x 3
3/8 x 16	3/8	16	1/2	8	1 x 3
3/8 x 18	3/8	18	1/2	7	1 x 3
3/8 x 20	3/8	20	1/2	7	1 x 3
1/2 x 6	1/2	6	1/2	13	1 x 4
1/2 x 8	1/2	8	1/2	10	1 x 4
1/2 x 10	1/2	10	1/2	9	1 x 4
1/2 x 12	1/2	12	1/2	8	1 x 4
1/2 x 14	1/2	14	1/2	7	1 x 4
1/2 x 16	1/2	16	1/2	7	1 x 4
1/2 x 18	1/2	18	1/2	6	1 x 4
1/2 x 20	1/2	20	1/2	6	1 x 4
5/8 x 6	5/8	6	1/2	10	1 x 4
5/8 x 8	5/8	8	1/2	9	1 x 4
5/8 x 10	5/8	10	1/2	7	1 x 4
5/8 x 12	5/8	12	1/2	6	1 x 4
5/8 x 14	5/8	14	1/2	6	1 x 4
5/8 x 16	5/8	16	1/2	5	1 x 4
5/8 x 20	5/8	20	1/2	5	1 x 4
3/4 x 6	3/4	6	5/8	11	1-1/8 x 5
3/4 x 8	3/4	8	5/8	9	1-1/8 x 5
3/4 x 10	3/4	10	5/8	7	1-1/8 x 5
3/4 x 12	3/4	12	5/8	6	1-1/8 x 5
3/4 x 14	3/4	14	5/8	6	1-1/8 x 5
3/4 x 16	3/4	16	5/8	6	1-1/8 x 5
3/4 x 18	3/4	18	5/8	5	1-1/8 x 5
3/4 x 20	3/4	20	5/8	5	1-1/8 x 5
1 x 6	1	6	3/4	7	1-1/4 x 5
1 x 8	1	8	3/4	6	1-1/4 x 5
1 x 10	1	10	3/4	5	1-1/4 x 5
1 x 12	1	12	3/4	4	1-1/4 x 5
1 x 14	1	14	3/4	4	1-1/4 x 5
1 x 16	1	16	3/4	3	1-1/4 x 5
1 x 18	1	18	3/4	3	1-1/4 x 5
1 x 20	1	20	3/4	3	1-1/4 x 5
1-1/4 x 6	1-1/4	6	13/16	4	1-1/2 x 5
1-1/4 x 8	1-1/4	8	13/16	4	1-1/2 x 5
1-1/4 x 10	1-1/4	10	13/16	3	1-1/2 x 5
1-1/4 x 12	1-1/4	12	13/16	3	1-1/2 x 5
1-1/4 x 14	1-1/4	14	13/16	2	1-1/2 x 5
1-1/4 x 16	1-1/4	16	13/16	2	1-1/2 x 5
1-1/2 x 6	1-1/2	6	7/8	4	1-3/4 x 5-1/2
1-1/2 x 8	1-1/2	8	7/8	3	1-3/4 x 5-1/2
1-1/2 x 10	1-1/2	10	7/8	3	1-3/4 x 5-1/2
1-1/2 x 12	1-1/2	12	7/8	2	1-3/4 x 5-1/2
1-1/2 x 14	1-1/2	14	7/8	2	1-3/4 x 5-1/2
1-1/2 x 16	1-1/2	16	7/8	2	1-3/4 x 5-1/2

NOTE: Requirements are not less than those shown except where otherwise indicated.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.

3.4.6 Controls. The controls shall consist of not less than the following:

3.4.6.1 Ram drive control. The ram drive control shall include a start-stop switch, necessary selector switches, and foot controls to provide for all functional operations of the machine. The selector switches, in conjunction with the foot controls, shall control the drive to perform the following functions:

a. "Single stroke" - The ram shall complete only one stroke regardless of how long the foot control is held down. When the foot control is released before the stroke has been completed the ram shall return to the top position and stop.

b. "Inch" - The ram shall operate along the normal stroke direction only as long as the foot control is held down.

3.4.6.2 Rake angle. Unless otherwise specified, either an adjustable rake angle or a fixed rake angle shall be furnished on machines of 1/2 inch capacity and less. When only one type rake angle is acceptable for machines of 1/2 inch and less, the rake angle required shall be as specified (see 6.2.1). Machines with a fixed rake angle shall comply with the rake angle requirements contained in table I. Machines of 5/8 inch capacity and greater shall be furnished with an adjustable rake angle.

3.4.6.2.1 Rake angle adjustment. Machines with an adjustable rake angle shall be provided with a rake angle adjustment control which indicates the rake angle range and setting. Maximum rake angle adjustment shall be not less than the requirements contained in table I. Unless otherwise specified, adjustable rake angle machines of 1/2 inch capacity and less shall have a manual means of obtaining rake angle adjustments. When specified (see 6.2.1), machines of 1/2 inch capacity and less shall have a power operated rake angle adjustment. Adjustable rake angle shearing machines of 5/8 inch capacity and greater shall have a power operated rake angle adjustment.

3.4.6.3 Ram stroke adjustment control. When specified (see 6.2.1), a means for controlling the length of the ram stroke shall be furnished. The ram stroke adjustment control shall provide an adjustable range of ram movement adequate for the machine to shear material of the rated capacity at the rated capacity rake angle setting specified in table I.

3.4.7 Electrical system. The electrical system shall conform to NFPA 79. Each machine shall be wired to draw all of its electrical power from a single 230/460 volt, 3 phase, 60 Hz circuit. Each machine shall have a fused safety disconnect switch or circuit breaker. The machine shall be initially wired for operation on 460 volts. An identified terminal for grounding the machine when it is installed shall be mounted in or near the

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.

disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in NFPA 79 for the disconnect fuse rating.

3.4.7.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Each motor enclosure shall meet the requirements for a dripproof enclosure. Mechanical motor starter relays may be used with the main drive motor.

3.4.7.2 Solid state components. Except for relays allowed in 3.4.7.1, solid state design shall be used throughout for electronic components.

3.4.7.3 Control circuit voltage. Auxiliary control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage no greater than 120 volts.

3.4.8 Gauges. The machine shall include a back gauge, front gauge, and two side squaring gauges. The gauges shall accommodate workpiece sizes up to the capacity of the machine.

3.4.8.1 Back gauge. A back gauge shall be furnished on the rear of the machine to provide an adjustable reference edge between the back edge of the workpiece and the lower shear knife. The gauge shall be adjustable for any position up to 30 inches for 1/4 inch capacity shears, and at least 36 inches for all other shears from the stationary shear knife cutting edge. After the workpiece is positioned against the gauge and secured to the table by the hold-downs, the back gauge shall permit an unrestricted drop of the sheared section. The back gauge shall be controlled by a push-button operated control unit with either an electric or hydraulic drive. The control unit shall be provided with a direct reading register type position indicator for indicating the distance between the lower shear knife and the gauge. The position indicator shall be located on the machine in a position that is readable from the operator's station. An adjustment shall be provided for correcting misalignment of the back gauge.

3.4.8.2 Front gauge. The front gauge shall consist of not less than two support arms with adjustable work stops, to provide an adjustable reference edge between the front edge of the workpiece and the lower shear knife. The work stops shall be adjustable for positioning the workpiece at any position from not more than six inches to not less than 48 inches from the lower shear knife.

3.4.8.3 Side squaring gauges. A side squaring gauge shall be provided on both sides of the table for positioning the side edge of the workpiece at right angles to the stationary shear knife. Calibrated measuring scales shall be recessed in either the side squaring gauge or the table top to aid in positioning the workpiece to the desired location.

3.4.8.4 Angle gauge. When specified (see 6.2.1), the machine shall be provided with an adjustable angle gauge for shearing workpieces at angles without the use of scribed lines.

Source: https://assist.dla.mil -- Downloaded: 202610-23T04:37Z
Check the source to verify that this is the current version before use.

3.4.8.5 Slitting gauge. When specified (see 6.2.1), the gap type shearing machine shall be provided with a gauge for guiding the workpiece in slitting operations.

3.4.8.6 Light beam shearing gauge. When specified (see 6.2.1), the machine shall be provided with a shearing gauge that utilizes light beam shadows at the shearing line. The light beam shearing gauge shall be wired through the electrical controls of the machine.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the machine shall be not less than those specified in table I for the machine size specified.

3.6 Performance. The machine shall be capable of shearing mild steel of not less than 60,000 pounds per square inch (psi) tensile strength in workpiece sizes ranging up to the rated capacity for the thickness and width as specified in table I. The rated capacity shall be sheared with the upper shear knife rake angle fixed, if applicable, or set at a rake angle not greater than that specified in table I. Shearing shall be accomplished with a shear knife clearance of not more than 15 percent of the workpiece thickness for shearing workpiece thicknesses up to 3/4 inch and not more than 25 percent of the workpiece thickness for shearing workpiece thicknesses above 3/4 inch. The sheared edge of the bed cut shall be square with the workpiece surface resting on the table within ±3 angular degrees. The workpiece shall be free of visible hold-down cylinder damage.

3.6.1 Shearing accuracy. The machine shall be capable of shearing 10 gauge material ranging up to the full width capacity of the machine to precision tolerances as follows:

a. Dimensions - The width of the sheared workpiece shall be within ± 0.005 inch of the back gauge setting.

b. Parallelism - The sheared cut shall be parallel to the sheared (trimmed) edge positioned against the back gauge within ±0.002 inch per linear foot of workpiece width.

c. Squareness - The sheared cut shall be square with a sheared (trimmed) edge positioned against either side gauge within ±0.002 inch per linear foot of workpiece width.

3.6.2 Additional performance requirements. Performance features or shearing accuracies required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.7 Standard equipment. All equipment normally furnished with the manufacturer's standard commercial shearing machine shall be furnished.

3.8 Optional equipment. When required, optional equipment shall be furnished as specified (see 6.2.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.

3.9 Repair parts. Repair parts shall be furnished as specified (see 6.2.1).

3.10 Markings on plates and charts. All words on plates and charts shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.10.1 Lubrication chart or plate. A lubrication chart or plate shall be securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder or permanently sealed between clear plastic sheets, with suitable means for mounting. The following information shall be furnished on the chart or plate:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.10.2 Nameplate. A nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix keyed to the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided that the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Shearing capacity, rated workpiece thickness and width at rated rake angle setting
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Hour meter. The machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of not more than one hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine. Upon reaching 99,999 hours, the meter readout shall automatically return to zero and continue to accumulate time.

3.12 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

Source: https://assist.dla.mil -- Downloaded: 202610-23T04:37Z
Check the source to verify that this is the current version before use.

3.13 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsiblity for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When first article inspection is required it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the first article to pass the examination and all tests shall be cause for rejection.

4.5 Quality conformance inspection. A quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), the quality conform-

Source: https://assist.dla.mil -- Downloaded: 202610-23T04:37Z
Check the source to verify that this is the current version before use.

ance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the test, and the inspection shall be cause for rejection.

4.6 Examination. The machine shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Operational test. The machine shall be operated at no load for not less than one hour. All controls, motors, adjusting mechanisms, gauges, and the hydraulic system shall be actuated and observed to determine compliance with the requirements specified herein.

4.7.2 Performance tests. The machine shall be tested for conformance with the requirements of 3.6 by shearing mild steel of 60,000 psi tensile strength. The instruments and methods used in performing the tests shall be at the option of the contractor, provided they accurately determine the shearing capacity of the machine.

4.7.2.1 Shearing capacity. The shearing capacity of the machine shall be verified by shearing workpiece sizes of not less than the rated capacity specified in table I.

4.7.2.2 Shearing accuracy. The shearing accuracy of the machine shall be verified by shearing 10 gauge material workpieces of not less than the rated width capacity of the machine. The machine shall perform dimension shearing, parallel shearing, and square shearing to comply with 3.6.1.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The machines covered by this specification are intended for use in metal squaring operations that are generally required in all phases of metal fabrication work operations. The gap type shearing machines are also intended, with optional attachments, for slitting, notching, and angular shearing operations.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

Source: https://assist.dla.mil -- Downloaded: 202610-23T04:37Z
Check the source to verify that this is the current version before use.

a. Title, number, and date of this specification.

b. Size of machine required (see 1.2 and table I).

c. First article, if required (see 3.1).

d. If machine design and construction is required in a specific dimensional system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Dual calibration (US and SI), if required (see 3.2.1.1.1).

g. Additional safety and health requirments, if required (see 3.2.5).

h. Lubrication system, if different (see 3.2.9).

i. Painting, if different (see 3.3.5).

j. If electromagnetic interference control is required, specify the equipment and subsystems class and the emission and susceptibility required (see 3.3.9).

k. If specific type end housing is required, state the required type (see 3.4.1).

l. Ball transfer arrangement, for shears with less than 1/2 inch capacity, if required (see 3.4.2).

m. Hydraulic system in conformance with JIC Standards, if required (see 3.4.3).

n. If a specific type ram hydraulic cylinder unit is required, state required type (see 3.4.3.1).

o. Holddown system independent control, if required (see 3.4.3.2).

p. If machine is required to have a specific type rake angle, state required type (see 3.4.6.2).

q. Power operated rake angle adjustment, if required (see 3.4.6.2.1).

r. Ram stroke adjustment control, if required (see 3.4.6.3).

s. Angle gauge, if required (see 3.4.8.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.

t. Slitting gauge, if required (see 3.4.8.5).

u. Light beam shearing gauge, if required (see 3.4.8.6).

v. Size and capacity, if different (see 3.5).

w. Specify additional performance and accuracy requirements (see 3.6.2).

x. Specify and fully describe optional equipment, if required (see 3.8).

y. Specify applicable repair parts, if required (see 3.9).

z. First article inspection, if different (see 4.4).

aa. Quality conformance inspection, if different (see 4.5).

bb. If preservation, packing, and marking in accordance with MIL-M-18058 is required, specify level of preservation and packing required (see 5.1).

6.2.2 Contract data requirements. Required technical data such as operator's manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the shearing machine itself. It does not require compliance with those OSHA requirements that concern the shearing machine in its operating environment, such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. OSHA limits the total hazard level of these hazards in the environment and does not limit the hazard level of individual machines in the environment. The procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional requirements that will integrate the new machine into its future operating environment. The machine shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial machine supplied to the commerical market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required because the guard configuration is dependent on the size and configuration of the workpiece. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1 (g).

Source: https://assist.dla.mil -- Downloaded: 202610-23T04:37Z
Check the source to verify that this is the current version before use.

6.5 Cross-reference of classification changes. The machine classifications are the same as the previous issue of this specification with the exception of the deletion of size 3/8 x 18 previously covered in MIL-S-80086C.

6.6 Subject term (key word) listing.

Cutting
Metal cutting
Metal working
Shearing
Slicing
Slitting
Trimming

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Preparing activity:
DLA - IP

(Project 3445-0102)

Review activities:
Air Force - 84
DLA- GS

User activities:
Army - SM
Navy - MC, YD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-80086D	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.